Embedded Microcontroller-Based Industrial Monitoring and Safety Control System Using Secure MQTT with Digital Twin Integration

Problem Statement

Industrial environments rely heavily on machines such as motors, pumps, and conveyor systems that operate continuously to maintain production processes. The performance and safety of these machines depend on several electrical, mechanical, and environmental parameters including voltage, current, temperature, humidity, vibration, and rotational speed. If these parameters are not properly monitored, abnormal conditions may remain undetected until they lead to equipment failure, production downtime, or safety hazards.

Conventional monitoring systems often depend on manual inspection or basic display units that only provide real-time readings without storing historical data. These systems lack remote monitoring capability and do not support automated responses during fault conditions. As a result, problems such as motor overheating, excessive load, abnormal vibration, or voltage fluctuations may not be detected in time to prevent damage.

Industrial facilities also face safety challenges such as fire hazards, equipment overload, and unauthorized access to restricted areas. Without an automated monitoring and response system, these situations can lead to accidents and operational losses.

Additionally, modern industrial systems require secure communication to prevent unauthorized access and data tampering. Traditional systems often lack encryption, authentication, and network isolation, making them vulnerable to cyber threats.

To overcome these issues, a smart and secure monitoring system is required that can continuously collect sensor data, transmit it wirelessly, provide real-time monitoring through a dashboard, and support digital twin simulation. The system should also log data for analysis, ensure secure communication, and automatically activate safety mechanisms when abnormal conditions occur.

Abstract

Industrial monitoring plays a crucial role in maintaining the reliability, efficiency, and safety of automated systems. Continuous observation of machine parameters allows industries to detect abnormal conditions early and take corrective actions before serious failures occur. This project presents the design and implementation of an IoT-based industrial monitoring and safety control system using embedded microcontrollers, wireless communication technologies, and secure protocols.

The proposed system uses microcontrollers such as STM32, ESP32, or NodeMCU to collect real-time data from multiple sensors including temperature, humidity, voltage, current, vibration, flame detection, and infrared sensors. These sensors monitor environmental conditions, electrical parameters, machine load conditions, and safety events within the industrial environment.

The collected data is transmitted through Wi-Fi using TCP/IP and MQTT communication protocols with TLS encryption (MQTTS) to ensure secure communication. Authentication mechanisms such as username/password and device ID verification are used to prevent unauthorized access. Data encryption ensures confidentiality and integrity of transmitted data.

The monitoring server implemented using MATLAB or Python records sensor data in daily log files and generates graphical plots stored as PNG images. A real-time monitoring dashboard allows operators to visualize sensor readings remotely.

A digital twin model implemented using Node-RED replicates real-time machine behavior using live MQTT data. This virtual model enables simulation and “what-if” analysis to predict system behavior under different fault conditions without affecting the physical system.

The system also implements automatic safety control mechanisms. In case of abnormal conditions such as overheating, overload, fire detection, or intruder detection, protective actions such as activating cooling systems, stopping motors, enabling load sharing, and triggering alerts are performed.

Hardware Components

Main controller options:

• STM32 / ESP32 / NodeMCU microcontroller board

Sensors:

1. Temperature Sensor – monitors machine temperature
2. Humidity Sensor – monitors environmental humidity
3. Voltage Sensor – measures supply voltage
4. Current Sensor – measures motor current consumption
5. Vibration Sensor – detects abnormal mechanical vibrations
6. Flame Sensor – detects fire hazards
7. IR Intruder Sensor – detects unauthorized entry
8. IR Speed Sensor – motor RPM measurement

Other components:

- Relay Module (motor and pump control)
- Buzzer (danger alert)
- Wi-Fi communication module (if required depending on controller)
- Motor and pump system
- Power supply
- Connecting wires and circuit components

Tools Used

Software Tools

- STM32CubeIDE / Arduino IDE
- MATLAB / Python Server
- Web Dashboard / IoT Platform
- Node-RED (Digital Twin)

• Graph plotting libraries

Communication Protocols

• TCP/IP

• MQTT over TLS (MQTTS)

Security Features

• Secure communication using MQTT over TLS (mqtts://)

• Authentication using username/password and device ID

• Data encryption using TLS/SSL

• Firewall and network isolation for industrial systems

• Data validation to detect abnormal or corrupted sensor values

Digital Twin Implementation

A digital twin is implemented using Node-RED. It receives real-time data via MQTT and mirrors the actual machine behavior. The Node-RED dashboard allows visualization and simulation of system parameters.

“What-if” simulations include:

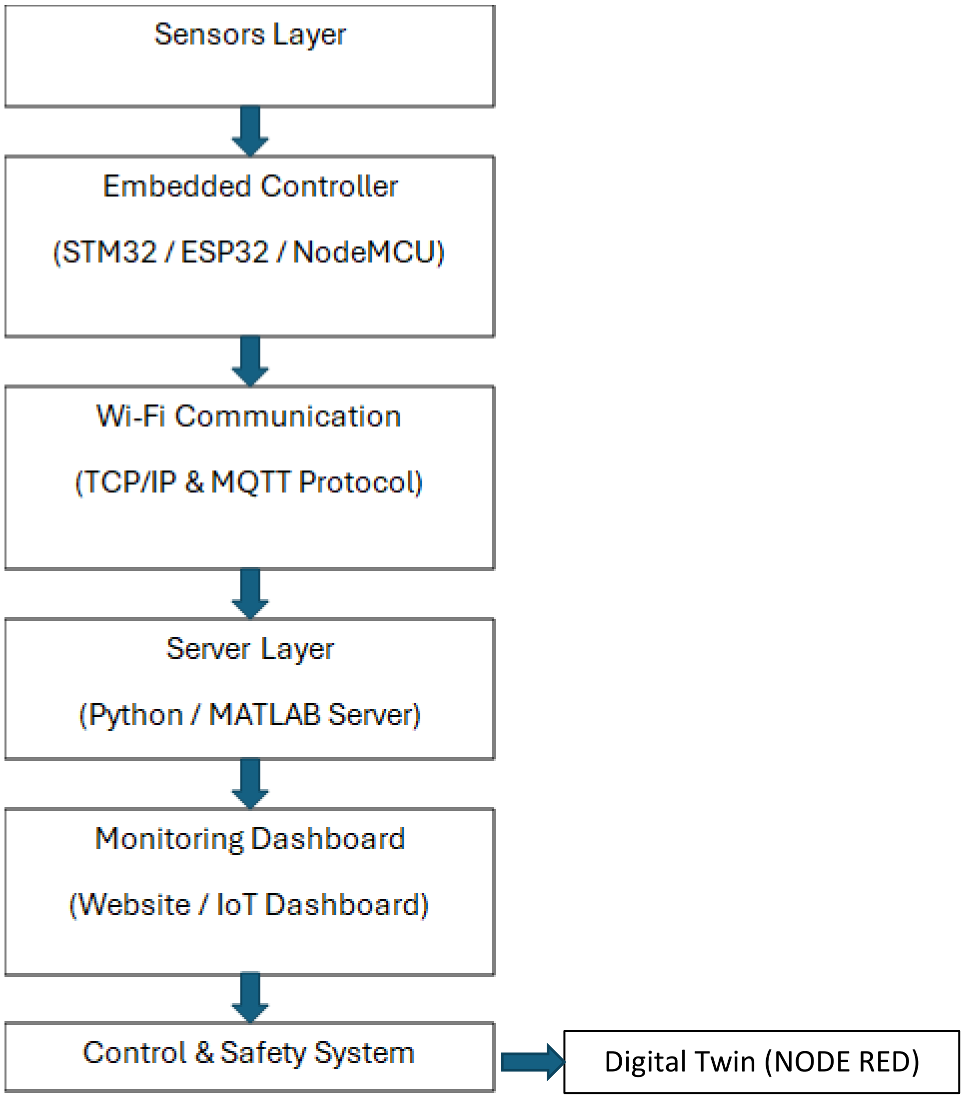
• Overheating scenarios

• Motor overload conditions

• Vibration faults

• Fire events

System Architecture



Outcome Parameters

Electrical parameters:

- Voltage
- Current
- Power consumption
- Load condition

Mechanical parameters:

- RPM of motor
- Conveyor belt load conditions
- Vibration level

Environmental parameters:

- Temperature
- Humidity

Safety parameters:

- Flame detection
- Intruder detection

These parameters are recorded in daily log files and visualized through graphs.

Expected Control Mechanisms

The system performs automatic actions based on sensor readings.

Temperature monitoring:

- If temperature exceeds threshold → warning
- Cooling system (fan or water tap) activated

Flame detection:

- Motor immediately turned OFF
- water pump ON
- Buzzer activated
- Danger notification sent to security

Intruder detection:

- IR sensor detects unauthorized access
- Alert notification generated

Electrical overload:

- If voltage/current/power exceeds safe limit → warning generated

Load sharing mechanism:

- If motor load increases due to high conveyor belt tension (RPM drop or Current increases)
- Secondary motor is activated through relay
- Load is shared between motors

Emergency alerts:

- Buzzer activated in critical conditions
- Notification sent through monitoring system

Every parameter and control operation will be displayed on the dashboard/ website.

Expected Outcome

The system will:

- Continuously monitor industrial equipment parameters
- Provide real-time monitoring through a dashboard
- Store sensor data in daily log files
- Generate graphical reports (PNG graphs) for analysis
- Detect abnormal conditions automatically
- Activate safety control mechanisms
- Improve industrial safety and equipment reliability
- Create historical data useful for future analysis
- Digital twin model using Node-RED
- “What-if” simulation capability
- Improved reliability and safety

Expected Cost

Component	Cost (INR)
Microcontroller (STM32/ESP32/NodeMCU)	₹400 – ₹2500
Sensors (8 sensors total)	₹1000 – ₹1500
Relay modules	₹200 – ₹400
Buzzer	₹50 – ₹100
Motor / pump	₹500 – ₹1500
Power supply & wiring	₹300 – ₹500

Total estimated project cost:

₹2500 – ₹5000 (approx.)

Future Scope

• Predictive maintenance using ML

• AI-based anomaly detection

• SCADA integration

• Energy optimization

• Advanced digital twin with AI

• Large-scale IIoT deployment